

Solving the Shortest Vector Problem in $2^{0.5n+o(n)}$ Time via Mid-point Hessian

Exact transport escapes and a response-floor obstruction

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Draft response to Hhan's mid-point Hessian algorithm

August 2026

Abstract

Hhan recently gave a randomized algorithm for exact shortest vector problem running in time $2^{0.6039n+o(n)}$ and space $2^{0.5n+o(n)}$. Its endpoint is a geometric fact about the Hessian of the periodic Gaussian: at the midpoint of a shortest vector, an extreme eigenvector points toward that shortest vector and can be completed by preprocessing bounded distance decoding.

We study the information available before the Walsh–Hadamard search over the 2^n parity midpoints. A single panel of full-dual discrete Gaussian samples gives the traceless periodic Hessian, and its first two derivatives, at every point of the continuous torus. We prove an exact half-grid stationarity and snapping lemma, a bounded narrowing-importance identity, and a uniform empirical derivative theorem using $2^{(0.4002+o(1))n}$ samples at a rigorously controlled smooth scale. A log-trace-exponential objective removes every eigengap assumption from the ascent vector field. These results give a black-box reduction: any robust, samplable ascent law reaching a shortest-vector parity with probability $2^{-o(n)}$ yields a half-exponential algorithm.

We then resolve the proposed universal Haar-seeded law negatively. For the rectangular family $L_n = \text{diag}(1, 2^n, \dots, 2^n)\mathbb{Z}^n$, the correct single scale is $\Theta(n^{-1/2})$. Outside exponentially small coordinate slabs, every third derivative driving soft-Hessian ascent is $\exp(-\Omega(n4^n))$. Hence any polynomial-step schedule of polynomial bit complexity leaves almost every Haar seed in its initial parity cell. The shortest parity is reached with probability at most 2^{-n+1} , and the robust basin is no larger. This disproves the manuscript's former Conjecture 13.1. We then give exact lattice-adapted repairs. Short dual vectors impose syndrome-zero equations on every shortest parity; the coefficient ellipsoid gives Hamming and spectral-tube covers; bounded coefficient energy admits exact generalized-distributive-law elimination; and an obtuse superbase reduces SVP to one global minimum cut. The last clause shows that the all-ones cancellation fan is submodular, rather than a genuine hard core.

For the surviving geometry we prove a causal transport formula. An infinitesimal isotropic displacement of a uniform point modulo the lattice crosses a facet of parity θ at rate proportional to that facet's area. The associated cone-volume weights form an exact isotropic frame, and every shortest facet retains a constant-area robust core with explicit inverse-polynomial reach and energy margin. A separate gap theorem gives a direct spherical winner cap. These statements reduce the unresolved case to a joint core in which every known cover, factor elimination, and submodular escape is expensive; the remaining geometric obligation is a lower bound on shortest direct mass plus shortest cone-volume mass.

We then prove that this geometric obligation would still not complete the proposed empirical line scan. Hhan's weighted shell estimate suppresses the non-colliding theta tail on a homothetic facet core only when the retained area is exponentially smaller than required. More decisively, at every tail-suppressed point whose theta response lies above the sampling noise floor, the tangential displacement from the midpoint is confined to an explicit Euclidean ball. Even if all $2^{n/2+o(n)}$ samples are spent on one query, that ball contains at most $2^{-(1.0289+o(1))n}$ of a shortest facet. Retaining $2^{-(0.0264828+o(1))n}$ would require sample exponent at least $0.61169\dots$. Thus polynomial weighted-tail suppression and a sufficient empirical theta-response floor cannot coexist on enough of the robust facet core within the half-exponential budget.

Independently of that hypothesis, we transfer the moving-stabilizer spherical-code bound of [6] through Hhan's shell estimates. Its one-row subfamily gives exponent 0.602470 , subject to the cited theorem and a one-dimensional numerical enclosure. Thus the note contains both a checkable update of Hhan's parameter ledger and a conditional route to the half exponent.

Conditioned on a replacement shortest-parity basin hypothesis for an adapted law with directly observable endpoints, we obtain a randomized classical algorithm for search-SVP in expected time and worst-case space $2^{0.5n+o(n)}$, up to polynomial factors in the input length. More generally, an attraction probability $2^{-(\delta+o(1))n}$ suffices for every $\delta < 0.02648$. Thus the note does not prove an unconditional half-exponent algorithm: it proves the analytic reduction, refutes the first universal candidate, and proves that the natural log-theta line detector cannot repair that candidate at the desired exponent. The surviving target must control response-weighted shortest collisions or use a different nonlocal estimator; direct-plus-cone mass alone is insufficient.

AI use disclosure. The technical route, finite experiments, and initial manuscript draft were developed with substantial assistance from OpenAI Codex. Every theorem whose proof is complete is separated from the explicitly labeled geometric hypothesis. The human author of any circulated version is responsible for checking the arguments, replacing the author placeholder, and deciding which claims are suitable for dissemination.

Contents

1	Introduction	3
1.1	Technical overview	3
1.2	Contributions and logical status	4
2	Preliminaries	5
2.1	Lattices, Gaussian parameters, and decoding	5
2.2	The target midpoint recovery lemma	6
3	A direct spherical-bound improvement	6
4	The continuous periodic-Hessian oracle	7
4.1	Traceless Fourier representation	7
4.2	Half-grid stationarity and exact parity snapping	8
5	Narrowing importance sampling	8
6	A uniform empirical derivative oracle	9
7	A gap-free soft Hessian objective	10
8	The samplable seed law	11
9	The geometric existence clause	11
10	The conditional $2^{n/2+o(n)}$ algorithm	12
11	Foundational consequences	13
11.1	Information lives on the whole torus	13
11.2	Scale transport is needed only for the observable	13
11.3	No eigengap is needed during search	13
11.4	All false candidates are one-sided errors	13
12	Finite evidence	13
12.1	Soft ascent on a cancellation family	13
12.2	Finite-population oracle checks	14

13 Resolution of the universal-basin conjecture	14
13.1 An exact lattice-adapted parity sieve	15
13.2 A spectral support-chain cover	16
13.3 Exact factor transports	17
13.4 A direct gap escape	18
13.5 Causal transport through the Voronoi tiling	19
13.6 A log-theta collision kernel	20
13.7 Robust shortest-facet cores	22
13.8 Tail suppression and the response-floor barrier	22
13.9 The remaining joint-core theorem	24
14 Conclusion	24

1 Introduction

Let $L = L(B) \subset \mathbb{R}^n$ be a full-rank lattice. The search shortest vector problem asks for a nonzero $v \in L$ of norm $\lambda_1(L)$. The best previously proved general classical algorithm ran in $2^{n+o(n)}$ time and space through discrete Gaussian sampling [2]. Hhan’s recent mid-point Hessian algorithm [4] reduces the classical running time to $2^{0.6039n+o(n)}$ while using $2^{n/2+o(n)}$ space.

The central observation of [4] is local. If v is shortest and $s = \Theta(\lambda_1(L)/\sqrt{n})$ is chosen correctly, the Hessian of the periodic Gaussian at $v/2$ is, up to a scalar identity matrix and small operator error, a positive multiple of vv^T . Its leading eigenvector is therefore inverse-polynomially close to $v/\|v\|$. The preprocessing BDD oracle of Aggarwal, Chen, Kumar, and Shen [1] then recovers v in subexponential query time.

The expensive part is finding the parity class of v in $L/2L$. Hhan uses matrix-valued Walsh–Hadamard transforms, random affine sublattice cosets, and importance sampling. The present note asks what can be obtained from the same samples without first committing to the discrete set $L/2L$. Hhan’s Remark 1.2 already anticipates that the new code and packing bounds of [6] lower his constant. Section 3 gives a conservative, reproducible transfer using only the explicit one-row spherical subfamily; the continuous route begins after that direct update.

1.1 Technical overview

For $s > 0$, put $\rho_s(x) = \exp(-\pi\|x\|^2/s^2)$ and

$$F_s(z) = \frac{\rho_s(L+z)}{\rho_s(L)} = \mathbb{E}_{X \sim D_{L^*, 1/s}} \left[e^{2\pi i \langle X, z \rangle} \right].$$

Differentiation gives the dual representation

$$\nabla^2 F_s(z) = -4\pi^2 \mathbb{E} [XX^T \cos(2\pi \langle X, z \rangle)].$$

The radial identity component is irrelevant to the eigenvectors. We remove it at the sample level and work with

$$\mathcal{T}_s(z) := \nabla^2 F_s(z) - \frac{\text{tr}(\nabla^2 F_s(z))}{n} I_n = -4\pi^2 \mathbb{E} [Q(X) \cos(2\pi \langle X, z \rangle)], \quad Q(X) := XX^T - \frac{\|X\|^2}{n} I_n. \quad (1)$$

Every target sample can be reused at an arbitrary $z \in \mathbb{R}^n/L$, and differentiating (1) supplies continuous gradients and Hessians of the matrix field.

There are two exact structural facts. First, every half-grid point $B\theta/2$, $\theta \in \mathbb{F}_2^n$, is stationary for every entry of $\mathcal{T}_s$ at every scale. Second, if θ is the coefficient parity of a shortest vector v , then $B\theta/2 = v/2 \bmod L$. Hence a continuous method need not transport a critical point between scales. It only needs to enter the rounding cell of the correct parity; it can then snap to the exact midpoint and invoke Hhan’s target-scale recovery theorem.

To obtain a smooth vector field even at eigenvalue collisions, define

$$\Psi_{t,d,\gamma}(z) := \frac{\sigma_{t,d}}{\gamma} \log \operatorname{tr} \exp \left(\frac{\gamma \mathcal{T}_t(z)}{\sigma_{t,d}} \right), \quad (2)$$

where $\gamma = \operatorname{poly}(n)$ and $\sigma_{t,d}$ is an explicit midpoint signal scale. The gradient of (2) is analytic everywhere. Haar initialization on the torus followed by a fixed discretization of this gradient flow and exact half-grid snapping is our seed law.

The scale used for ascent is below the direct sampling threshold. We draw samples at a target parameter $r > t_0$ and narrow them to $t < r$ using weights bounded by one. For

$$r = 0.23675858, \quad t = 0.16,$$

the auxiliary second-moment parameter and importance exponent are

$$q = \frac{tr}{2r-t} = 0.1208271\dots > \frac{t_0}{2}, \quad \iota(t, r) = \frac{1}{2} \log_2 \frac{r^2}{t(2r-t)} = 0.0801086\dots \quad (3)$$

Thus the uniformly accurate smooth oracle uses

$$2^{(2t+\iota(t,r)+o(1))n} = 2^{(0.4001086+o(1))n} \quad (4)$$

samples. The target midpoint estimator uses $2^{(2r+o(1))n} = 2^{(0.4735172+o(1))n}$ samples. Both fit below the $2^{n/2+o(n)}$ discrete-Gaussian and preprocessing-BDD cost.

1.2 Contributions and logical status

The following components are proved in this note.

1. The moving-stabilizer spherical-code bound substitutes monotonically into Hhan's shell argument; its one-row specialization gives exponent below 0.602470 after the displayed numerical enclosure.
2. The full-torus traceless Hessian and all fixed-order physical derivatives are expectations of explicit matrix trigonometric polynomials.
3. Every half-grid point is stationary at every scale, and rounding an endpoint to the correct parity produces exactly the shortest midpoint modulo L .
4. Narrowing from r to t uses weights in $[0, 1]$. Its exact relative second moment has exponent $\iota(t, r)$ whenever the auxiliary parameter $q = tr/(2r-t)$ lies above $t_0/2$.
5. One importance-weighted sample panel uniformly estimates $\mathcal{T}_t$ and its first two derivatives over the entire torus. The uniform event permits adaptive queries without a pathwise union bound.
6. Log-trace-exponential smoothing transfers the matrix estimates to a gap-free scalar ascent field.
7. A robust shortest-parity basin lower bound implies a $2^{n/2+o(n)}$ -time and space algorithm by exact snapping, target-scale midpoint estimation, preprocessing BDD, and verification.
8. The universal Haar-seeded soft-Hessian law does not have the required basin mass. An anisotropic rectangular family gives success probability at most 2^{-n+1} for every polynomial-step, polynomial-description schedule.
9. Short-dual syndrome equations, a coefficient Hamming bound, and a stable spectral-tube lifting decomposition give unconditional lattice-adapted parity covers. Whenever their displayed entropy is $o(n)$, they instantiate the half-exponential reduction without an ascent hypothesis.
10. Bounded coefficient energy has an exact weighted-junction-width factorization. A displayed elimination order of width below the query allowance solves that input by min-sum elimination and backward preimage.

11. If a basis extends to an obtuse superbase, a discrete coarea identity reduces exact SVP to a global minimum cut. In particular the previously used simplex-cancellation family is polynomial-time solvable.
12. Infinitesimal causal transport through the Voronoi tiling has an exact facet-area law. Its cone-volume weights obey an isotropic matrix conservation identity, and shortest facets have explicit robust cores.
13. The centered Hessian of the log theta function is exactly a lattice-site posterior covariance. At a two-site facet collision it exposes $vv^T/4$ independently of tangential position, reducing line detection to explicit tail and relative-response estimates.
14. Those two estimates obey an incompatible exponent ledger. Weighted shell suppression on a homothetic core forces exponential area loss, while the empirical response floor confines accessible points to a ball occupying at most $2^{-(1.0289+o(1))n}$ of a shortest facet at the full $2^{n/2+o(n)}$ sample budget. Retaining exponent 0.0264828 would require sample exponent at least 0.61169...
15. A noncollinear length gap gives an explicit winner cap. The query ledger closes from this clause alone once the gap ratio exceeds 5.2670066...

The counterexample separates the proved reduction from its failed first instantiation. Hypothesis 9.2 records the condition needed by the reduction for the stated law; Theorem 13.1 proves that condition false. The lattice-adapted covers and exact factor transports below remove this obstruction on a broad solvable class. A universal half-exponential algorithm therefore requires more than a direct-plus-cone mass theorem on the residual joint core: it needs response-weighted collision mass or a different observable whose empirical error is not divided by the local theta value.

2 Preliminaries

We follow the notation and normalization of [4]. All logarithms are natural unless a base is shown. The dimension tends to infinity, and $o(1)$ terms are uniform over fixed parameters in the stated ranges. The input basis is rational and has polynomial bit length.

2.1 Lattices, Gaussian parameters, and decoding

For a nonsingular $B \in \mathbb{R}^{n \times n}$, let $L = B\mathbb{Z}^n$ and $L^* = B^{-T}\mathbb{Z}^n$. Write $\lambda = \lambda_1(L)$. As in [4], LLL reduction and a polynomial scale grid permit us to assume that a value d is available with

$$\lambda \leq d \leq (1 + 1/n)\lambda. \quad (5)$$

Let $\beta_0 = 2^{0.4014\dots}$ be the lattice-point counting constant used in [4], and define

$$\xi_a(d) := \sqrt{\frac{4na \ln 2}{\pi d^2}}, \quad t_0 := \frac{\beta_0^2}{4e \ln 2} = 0.23147\dots \quad (6)$$

We use two black-box results exactly as in Hhan's algorithm.

Theorem 2.1 (Above-smoothing DGS, [2, 4]). *If $a > t_0$ is fixed, one call at parameter $\xi_a(d)$ outputs up to $2^{n/2}$ samples jointly negligibly close to independent samples from $D_{L^*, \xi_a(d)}$ in time and space $2^{n/2+o(n)}$.*

Theorem 2.2 (Preprocessing BDD, [1, 4]). *There is randomized preprocessing using time and space $2^{n/2+o(n)}$ such that, after successful preprocessing, inverse-polynomial BDD queries take $2^{o(n)}$ time. Outputs on targets outside the decoding promise may be arbitrary, but exact lattice membership and norm can be verified.*

The last verification clause is useful: every wrong parity or unstable ascent path is harmless except for its running time.

2.2 The target midpoint recovery lemma

For $a > 0$, abbreviate

$$F_a(z) := F_{1/\xi_a(d)}(z), \quad H_a(z) := \nabla^2 F_a(z).$$

For $\theta \in \mathbb{F}_2^n$, choose its lift in $\{0, 1\}^n$. Hhan defines $G_{a,d}(\theta) = H_a(B\theta/2)$ and adds a scalar identity to obtain a positive semidefinite matrix $A_{a,d}(\theta)$. The following packages Lemmas 3.4–3.6 of [4] in the form used here.

Theorem 2.3 (Midpoint recovery, [4]). *Fix $t_0 < r < 1/4$ and suppose (5) holds. Let v be shortest and $v \in B\theta + 2L$. There is an explicit*

$$\mu_{r,d} = \xi_r^4 \lambda^2 2^{-rn+O(1)} \quad (7)$$

such that, after addition of a scalar identity,

$$H_r(B\theta/2) = \mu_{r,d} \frac{vv^T}{\lambda^2} + E + aI_n, \quad \|E\|_{\text{op}} \leq \mu_{r,d} 2^{-\Omega(n)}. \quad (8)$$

If an estimator of $H_r(B\theta/2)$ has operator error $O(\mu_{r,d}/n)$, then a leading eigenvector is $O(n^{-1/2})$ -close, up to sign, to v/λ . A preprocessing BDD query at the corresponding scaled direction returns $\pm v$ for all sufficiently large n .

Because adding or subtracting a scalar identity does not change eigenspaces, Theorem 2.3 remains valid for the traceless Hessian used below.

3 A direct spherical-bound improvement

Hhan's lattice-point estimates use the cap-optimized Kabatianskii–Levenshtein spherical-code exponent in two places: the global constant $\log_2 \beta$ and the shell function K_2 . The binary-code theorem of [6] does not itself apply to these Euclidean directions, but the spherical theorem in the same work does. We record the substitution because it yields a lower constant without the geometric hypothesis used later.

For $u \geq 0$, put

$$H_{\text{sph}}(u) = (1+u) \log_2(1+u) - u \log_2 u, \quad H_{\text{sph}}(0) = 0,$$

and for $a > b \geq 0$ put

$$\Gamma_{\text{row}}(a, b) = \frac{(a-b)(1+a+b)}{(1+2a)\sqrt{a(1+a)}}. \quad (\text{S1})$$

The moving one-row theorem of [6, Chapter 2] gives the valid whole-sphere exponent

$$B_{\text{row}}^{\text{dir}}(s) := \inf_{\substack{a > b \geq 0 \\ 2\Gamma_{\text{row}}(a,b) \geq s}} (H_{\text{sph}}(a) - H_{\text{sph}}(b)), \quad (\text{S2})$$

and its cap optimization can only decrease this value. Define

$$b_{\text{row}} := B_{\text{row}}^{\text{dir}}(1/2), \quad t_{0,\text{row}} := \frac{2^{2b_{\text{row}}}}{4e \ln 2}, \quad (\text{S3})$$

$$K_{\text{row}}(x) := \begin{cases} 0, & 1 \leq x \leq \sqrt{2}, \\ B_{\text{row}}^{\text{dir}}(1 - 2/x^2), & x > \sqrt{2}, \end{cases} \quad (\text{S4})$$

$$g_{2,\text{row}}(R) := 1 - \sup_{x \geq 1} \left(\min\{b_{\text{row}} + \log_2 x, 1 + K_{\text{row}}(x)\} - \frac{Rx^2}{2} \right). \quad (\text{S5})$$

For $g_{\text{row}}(r) = \frac{1}{2} \log_2(r/t_{0,\text{row}})$ and $\iota(r, R) = \frac{1}{2} \log_2(R^2/[r(2R-r)])$, Hhan's classical exponent ledger becomes

$$\chi(r, R) = \min\{g_{2,\text{row}}(R), \frac{1}{2} + g_{\text{row}}(r), 1 + \min(g_{\text{row}}(r), 2g_{\text{row}}(r)) - 2r, \frac{1}{2}\}, \quad (\text{S6})$$

$$c(r, R) = \max\{\frac{1}{2}, 2r + \iota(r, R), 1 - \chi(r, R)\}. \quad (\text{S7})$$

Theorem 3.1 (Moving-row transfer, numerical form). *Assume the spherical-code theorem of [6, Chapter 2] and the numerical enclosure (S8). Hhan’s classical mid-point Hessian algorithm has time complexity*

$$2^{0.602470n+o(n)}$$

and space complexity $2^{n/2+o(n)}$. More precisely, direct substitution of

$$r = 0.22189254, \quad R = 0.39935521$$

in (S2)–(S7) gives

$$\begin{aligned} b_{\text{row}} &= 0.39730560\dots, \quad t_{0,\text{row}} = 0.23015577\dots, \quad \chi = 0.39753095\dots, \\ 2r + \iota(r, R) &= 0.60246910\dots, \quad 1 - \chi = 0.60246905\dots \end{aligned} \quad (\text{S8})$$

Proof. Replace the spherical-code exponent in the shell count underlying Hhan’s Lemma 2.1 by (S2). This changes $\log_2 \beta$ to b_{row} and therefore changes the Gaussian threshold to (S3). In Hhan’s refined shell bound, replace the occurrence of the classical exponent at angle $1 - 2/x^2$ by (S4). Both replacements are pointwise upper bounds, so every subsequent Gaussian-mass, affine-coset, importance-sampling, and concentration inequality is unchanged except for the scalar functions in (S3)–(S7). The original optimization proof therefore gives exponent $c(r, R)$.

The two variational problems in (S2) and (S5) are one-dimensional after solving $2\Gamma_{\text{row}}(a, b) = s$ for a . Evaluating them at the displayed parameters gives (S8); the gap to 0.602470 is more than $9 \cdot 10^{-7}$. The nested formulas are evaluated directly rather than replaced by the binary-code exponent. Hence $c(r, R) < 0.602470$. Hhan’s space argument is unaffected and remains $2^{n/2+o(n)}$. $\square$

Remark 3.2. Theorem 3.1 is logically independent of the continuous basin hypothesis below. It is a direct improvement of Hhan’s proved algorithm if the cited spherical-code theorem and the numerical enclosure in (S8) are accepted. The full stabilizer hierarchy of [6] is pointwise no worse than the one-row family, but no additional numerical improvement from that hierarchy is claimed here.

4 The continuous periodic-Hessian oracle

4.1 Traceless Fourier representation

For $X \in \mathbb{R}^n$, define the traceless quadratic matrix

$$Q(X) := XX^T - \frac{\|X\|^2}{n} I_n. \quad (9)$$

For a scale $a > 0$, define

$$\mathcal{T}_a(z) := H_a(z) - \frac{\text{tr } H_a(z)}{n} I_n. \quad (10)$$

Lemma 4.1 (Full-torus oracle). *For every $z \in \mathbb{R}^n$,*

$$\mathcal{T}_a(z) = -4\pi^2 \mathbb{E}_{X \sim D_{L^*, \varepsilon_a}} [Q(X) \cos(2\pi \langle X, z \rangle)]. \quad (11)$$

For every fixed integer $j \geq 0$ and directions $h_1, \dots, h_j \in \mathbb{R}^n$,

$$D^j \mathcal{T}_a(z)[h_1, \dots, h_j] = -4\pi^2 (2\pi)^j \mathbb{E} \left[Q(X) \prod_{m=1}^j \langle X, h_m \rangle \cos \left(2\pi \langle X, z \rangle + \frac{j\pi}{2} \right) \right]. \quad (12)$$

In particular, one sample panel gives unbiased estimators at arbitrary, adaptively selected torus points.

Proof. Poisson summation gives $F_a(z) = \mathbb{E}[e^{2\pi i \langle X, z \rangle}]$. The distribution is symmetric under $X \mapsto -X$, so the imaginary part vanishes. Differentiating twice and subtracting the normalized trace gives (11). Further directional differentiation uses $d^j \cos u / du^j = \cos(u + j\pi/2)$ and proves (12). The final statement follows because z appears only in the trigonometric factor; the sample distribution does not depend on the query. $\square$

The traceless projection is not cosmetic. It removes the radial identity fluctuation from every individual sample while preserving every eigenvector and eigengap of $H_a(z)$.

4.2 Half-grid stationarity and exact parity snapping

The next fact does not require a Gaussian estimate or a spectral gap.

Lemma 4.2 (Scale-uniform half-grid stationarity). *For every $a > 0$ and $\theta \in \mathbb{Z}^n$,*

$$D\mathcal{T}_a(B\theta/2) = 0. \quad (13)$$

Consequently, every differentiable spectral function of $\mathcal{T}_a$, including the soft maximum in Section 7, is stationary at $B\theta/2$.

Proof. Write $X = B^{-T}k$ with $k \in \mathbb{Z}^n$. At $z = B\theta/2$, $2\pi\langle X, z \rangle = \pi k^T \theta$, and every sine factor in the first derivative of (11) vanishes. The chain rule proves the second claim. $\square$

Lemma 4.3 (Parity snapping). *Let $c \in \mathbb{R}^n$ and suppose there is a $\theta \in \mathbb{Z}^n$ with*

$$\|2c - \theta\|_\infty < \frac{1}{2}. \quad (14)$$

Then coordinate rounding of $2c$ recovers $\theta \bmod 2$. If a shortest vector v satisfies $v = B\theta + 2w$ for some $w \in L$, then

$$B\theta/2 = v/2 \pmod{L}. \quad (15)$$

Therefore $\mathcal{T}_r(B\theta/2) = \mathcal{T}_r(v/2)$.

Proof. Condition (14) gives unique coordinate rounding. From $v = B\theta + 2w$, divide by two to obtain $B\theta/2 = v/2 - w$. Periodicity of F_r and its derivatives gives the last claim. $\square$

Thus a smooth-scale critical point is not transported to the target scale. Its parity is transported algebraically by one exact rounding operation.

5 Narrowing importance sampling

Fix a source parameter $r > t_0$ and a smaller ascent parameter $t < r$. Put $\xi_a = \xi_a(d)$ and let P_a denote D_{L^*, ξ_a} . Since $\xi_t < \xi_r$, define

$$u_t(x) := \frac{\rho_{\xi_t}(x)}{\rho_{\xi_r}(x)} = \exp \left[-\pi \|x\|^2 \left(\frac{1}{\xi_t^2} - \frac{1}{\xi_r^2} \right) \right]. \quad (16)$$

The elementary but useful point is

$$0 < u_t(x) \leq 1. \quad (17)$$

There is no upper-tail problem in the importance weight; the loss comes from its small normalizing mean.

Lemma 5.1 (Exact narrowing identity). *For every integrable matrix-valued A ,*

$$\mathbb{E}_t[A(X)] = \frac{\mathbb{E}_r[u_t(X)A(X)]}{\mathbb{E}_r[u_t(X)]}. \quad (18)$$

Let

$$q = q(t, r) := \frac{tr}{2r - t}, \quad \iota(t, r) := \frac{1}{2} \log_2 \frac{r^2}{t(2r - t)}. \quad (19)$$

Then

$$\frac{\mathbb{E}_r[u_t(X)^2]}{\mathbb{E}_r[u_t(X)]^2} = 2^{\iota(t, r)n} \frac{\rho_{1/\xi_q}(L) \rho_{1/\xi_r}(L)}{\rho_{1/\xi_t}(L)^2}. \quad (20)$$

If $q > t_0/2$, the final theta-mass ratio is $1 + 2^{-\Omega(n)}$ and the relative second moment is $2^{(\iota(t, r) + o(1))n}$.

Proof. Expanding both expectations in (18) cancels ρ_{ξ_r} and gives the P_t expectation after division by the mass ratio. Moreover,

$$\frac{\rho_{\xi_t}(x)^2}{\rho_{\xi_r}(x)} = \rho_{\xi_q}(x)$$

because $1/q = 2/t - 1/r$. It follows that the left side of (20) is

$$\frac{\rho_{\xi_q}(L^*)\rho_{\xi_r}(L^*)}{\rho_{\xi_t}(L^*)^2}.$$

Poisson summation changes the power-of-scale factor into $(qr/t^2)^{n/2} = 2^{\iota(t,r)n}$ and leaves the displayed primal theta-mass ratio. Hhan's Gaussian mass bound implies $\rho_{1/\xi_a}(L) = 1 + 2^{-\Omega(n)}$ for every fixed $a > t_0/2$. Since $q < t < r$, the assumption on q proves the last claim. $\square$

6 A uniform empirical derivative oracle

We now show that adaptivity does not require fresh samples. All derivatives below are physical derivatives on $\mathbb{R}^n$, not coefficient derivatives; this avoids introducing the condition number of the input basis into the random summands.

Let

$$\bar{\mu}_{t,d} := \xi_t^4 d^2 2^{-tn}. \quad (21)$$

For the correct length guess, this is the coefficient scale of the shortest-pair contribution, up to polynomial and constant factors. It does not assert that this rank-one contribution dominates the full Hessian when $t < t_0$; that geometric issue is precisely what the basin hypothesis below must control.

Given independent $X_1, \dots, X_M \sim P_r$, define the ratio estimator

$$\widehat{\mathcal{T}}_t(z) := -4\pi^2 \frac{\sum_{i=1}^M u_t(X_i) Q(X_i) \cos(2\pi \langle X_i, z \rangle) \mathbf{1}_{\mathcal{G}_i}}{\sum_{i=1}^M u_t(X_i) \mathbf{1}_{\mathcal{G}_i}}, \quad (22)$$

where $\mathcal{G}_i$ is the event $\|X_i\| \leq C\xi_r n$ for a sufficiently large fixed C . Its derivative estimators are obtained by differentiating the trigonometric factor in (22).

Theorem 6.1 (Uniform narrowing oracle). *Fix constants r, t with*

$$t_0 < r < 1/4, \quad 0 < t < r, \quad q(t, r) > t_0/2. \quad (23)$$

For every fixed $A > 0$, there is a fixed C_A such that, with

$$M = (n + \text{size}(B))^{C_A} 2^{(2t+\iota(t,r))n}, \quad (24)$$

the following holds with probability at least $1 - \exp(-n^2)$: simultaneously for every $z \in \mathbb{R}^n/L$ and $j \in \{0, 1, 2\}$,

$$\sup_{\|h_1\|, \dots, \|h_j\| \leq 1} \|D^j \widehat{\mathcal{T}}_t(z)[h_1, \dots, h_j] - D^j \mathcal{T}_t(z)[h_1, \dots, h_j]\|_{\text{op}} \leq \frac{\bar{\mu}_{t,d} \xi_t^j}{n^A}. \quad (25)$$

The same sample panel supports an arbitrary adaptive sequence of queries. For ideal independent samples the failure probability is at most $\exp(-n^2)$. For the actual DGS output it is at most $2^{-\Omega(n)}$ after the statistical-distance loss.

Proof. We expose the exponent calculation and then account for uniformity. For a fixed z , directions h_m , and matrix entry, the numerator summand for the j th derivative is bounded on $\mathcal{G}_i$ by

$$(C' \xi_r n)^{2+j}. \quad (26)$$

The omitted Gaussian tail contributes at most $\bar{\mu}_{t,d} \xi_t^j / n^{A+2}$ after increasing C ; this is the standard discrete-Gaussian tail integration used in [4].

Let $m_t = \mathbb{E}_r[u_t(X)]$. By Lemma 5.1,

$$\frac{\mathbb{E}_r[u_t^2]}{m_t^2} = 2^{(\iota(t,r)+o(1))n}. \quad (27)$$

Scalar Bernstein concentration applied to the bounded denominator gives $\sum_i u_t(X_i) = (1 + o(1))Mm_t$. Matrix Bernstein applied to the centered weighted numerator, together with (27), has the same variance as $M2^{-(\iota+o(1))n}$ target samples. The target signal scale is 2^{-tn} up to polynomial factors, so inverse-polynomial relative accuracy requires $2^{(2t+\iota)n}$ samples. The polynomial n^{C_A} makes the failure probability at a fixed query smaller than $\exp(-n^{C'})$ for an arbitrarily large fixed C' .

It remains to make the event uniform. On $\mathcal{G}_i$, (12) bounds the next physical derivative of every summand by a number whose logarithm is polynomial in n and the input bit length. A rational fundamental parallelepiped of L has diameter at most exponential in that bit length. Consequently it has a net fine enough for (25) whose logarithmic cardinality is polynomial in n and the input bit length. Apply the fixed-query bound to the net, to a fixed operator-norm net on the unit sphere, and to $j = 0, 1, 2$; choose C' above the resulting polynomial; and use the next derivative bound to extend from the net to the whole torus. This proves (25). Since the event already quantifies over every z , adaptively chosen queries require no additional union bound. The statistical distance in the DGS theorem adds at most $2^{-\Omega(n)}$ to the ideal-sample failure probability. $\square$

Remark 6.2. The theorem is a statement about oracle accuracy, not automatic trajectory shadowing. Near a separatrix, an arbitrarily small perturbation can change the terminal basin. The robust basin hypothesis in Section 9 quantifies the dynamical margin that is additionally required.

7 A gap-free soft Hessian objective

The map $T \mapsto \lambda_{\max}(T)$ is not differentiable when the leading eigenvalue is multiple. This is inconvenient precisely at symmetric lattices. For $\gamma > 0$ and $\sigma = \bar{\mu}_{t,d}$, define

$$\psi_{\gamma,\sigma}(T) := \frac{\sigma}{\gamma} \log \text{tr} e^{\gamma T/\sigma}, \quad \Psi_{t,d,\gamma}(z) := \psi_{\gamma,\sigma}(\mathcal{T}_t(z)). \quad (28)$$

Lemma 7.1 (Soft spectral calculus). *For every symmetric T ,*

$$\lambda_{\max}(T) \leq \psi_{\gamma,\sigma}(T) \leq \lambda_{\max}(T) + \frac{\sigma \log n}{\gamma}. \quad (29)$$

If

$$P_T := \frac{e^{\gamma T/\sigma}}{\text{tr} e^{\gamma T/\sigma}},$$

then

$$D\psi_{\gamma,\sigma}(T)[E] = \text{tr}(P_T E). \quad (30)$$

Every derivative of ψ is analytic for every symmetric T , and the operator norm of its second Fréchet derivative is at most $O(\gamma/\sigma)$. Therefore Theorem 6.1, with a larger polynomial factor, uniformly estimates Ψ and $\nabla\Psi$ to inverse-polynomial absolute error without any eigengap assumption.

Proof. If $\lambda_1 \geq \dots \geq \lambda_n$ are the eigenvalues, then $e^{\gamma\lambda_1/\sigma} \leq \sum_i e^{\gamma\lambda_i/\sigma} \leq ne^{\gamma\lambda_1/\sigma}$, proving (29). The Fréchet derivative of $\log \text{tr} e^T$ gives (30). Matrix exponentiation and trace are analytic, and the trace exponential is strictly positive. The derivative in (30) has operator norm at most one, while the covariance formula for its derivative is bounded by $O(\gamma/\sigma)$. The physical first derivatives of $\mathcal{T}_t$ have polynomial norm by (12) and the Gaussian moment bounds. Consequently an error σ/n^A in $\mathcal{T}_t$ and its first derivative induces inverse-polynomial absolute error in Ψ and its gradient after increasing A . This is exactly the uniform event of Theorem 6.1. $\square$

When a polynomial eigengap is present and $\gamma = \text{poly}(n)$ is sufficiently large, P_T is exponentially close to the projector onto the leading eigenvector. When no gap is present, (30) remains a canonical convex mixture of the tied eigenspaces rather than an undefined branch choice.

8 The samplable seed law

Fix $t = 0.16$, $\gamma = n^2$, a polynomial step count $S = n^{C_0}$, and a deterministic step schedule $\eta_0, \dots, \eta_{S-1}$ of polynomial bit complexity. Let $\widehat{\Psi}$ be the empirical soft objective from the sample panel in Theorem 6.1. Define a bounded vector field

$$\widehat{V}(z) := \frac{d^4 \nabla \widehat{\Psi}(z)}{1 + d^3 \|\nabla \widehat{\Psi}(z)\|}. \quad (31)$$

The normalization is globally Lipschitz and keeps the step bounded; any fixed smooth normalization with polynomial Lipschitz bounds could be used instead. The powers of d make the field scale-equivariant: replacing L by αL replaces V by αV .

Soft-Hessian seed law $\widehat{\nu}_{L,d}$.

1. Draw $U \sim \text{Unif}([0, 1]^n)$ and put $Z_0 = BU \bmod L$.
2. For $s = 0, \dots, S - 1$, set

$$Z_{s+1} = Z_s + \eta_s \widehat{V}(Z_s) \pmod{L}.$$

3. Compute $c = B^{-1}Z_S$ modulo $\mathbb{Z}^n$, set $\theta = \text{round}(2c) \bmod 2$, and output $B\theta/2 \bmod L$.

This is a samplable probability law using one target DGS panel. It does not presuppose the shortest vector or its parity. Its exact counterpart $\nu_{L,d}$ replaces $\widehat{\Psi}$ by Ψ . All transcendental evaluations are performed to inverse-polynomial precision; the robustness margin below absorbs that deterministic approximation.

9 The geometric existence clause

Let $\Theta_{\min} \subseteq \mathbb{F}_2^n$ be the set of parities represented by shortest vectors. We need robustness because uniform oracle accuracy cannot stabilize a point lying exactly on a basin boundary.

Definition 9.1 (Robust shortest-parity basin). For $\varepsilon > 0$, let $\mathcal{S}_{L,d}(\varepsilon)$ be the set of Haar seeds Z_0 with the following property: every vector field $\widetilde{V}$ whose value is uniformly within εd of the exact normalized soft field V produces, under the fixed S -step schedule, a terminal point satisfying (14) with a parity in $\Theta_{\min}$ and with rounding margin at least ε . Set

$$p_{L,d}(\varepsilon) := \text{Haar}_{\mathbb{R}^n/L}(\mathcal{S}_{L,d}(\varepsilon)). \quad (32)$$

Hypothesis 9.2 (Robust shortest-parity basin). *There is a universal computable function $\kappa(n) = o(n)$ such that, for every full-rank lattice L , at one scale guess d satisfying (5), there is an inverse-polynomial $\varepsilon = n^{-C_1}$ for which*

$$p_{L,d}(\varepsilon) \geq 2^{-\kappa(n)}. \quad (33)$$

The step count, smoothing parameter, and constants C_0, C_1 are universal.

Hypothesis 9.2 is exactly the additional statement used by the $2^{n/2+o(n)}$ reduction for this candidate law. A stronger uniform-attraction theorem would not be needed, nor would a bound on the total number of local maxima. However, Theorem 13.1 below shows that the hypothesis itself is false for the universal Haar-seeded, single-scale law.

Proposition 9.3 (Empirical preservation of robust mass). *Assume Hypothesis 9.2. Increase the polynomial factor in Theorem 6.1 so that the normalized empirical vector field is within εd . Then a sample from $\widehat{\nu}_{L,d}$ snaps to $\Theta_{\min}$ with probability $2^{-\kappa(n)} - 2^{-\Omega(n)}$.*

Proof. On the uniform event of Theorem 6.1, Lemma 7.1 and the normalization in (31) put $\widehat{V}$ in the perturbation class quantified in the preceding definition. Every seed in $\mathcal{S}_{L,d}(\varepsilon)$ therefore snaps to a shortest parity. The exceptional oracle event for the actual DGS panel has probability at most $2^{-\Omega(n)}$. $\square$

10 The conditional $2^{n/2+o(n)}$ algorithm

We can now state the algorithm in the same verified-output model used by Hhan.

Algorithm 1: Continuous mid-point Hessian SVP.

1. Apply LLL, construct the polynomial grid of guesses d , and construct the preprocessing $n^{-1/3}$ -BDD data.
2. Fix $r = 0.23675858$ and $t = 0.16$. At each d , call above-smoothing DGS on L^* at $\xi_r(d)$ and retain enough samples for the smooth oracle and target midpoint estimators. Abort a scale returning too few samples.
3. Using the same panel and the weights (16), draw $K = \Theta(2^{\kappa(n)}) = 2^{o(n)}$ independent outputs of the soft-Hessian seed law $\widehat{v}_{L,d}$.
4. For every snapped parity θ , estimate the target traceless Hessian $\mathcal{T}_r(B\theta/2)$ from $n^C 2^{2rn}$ unweighted source samples. Compute a leading eigenvector q and query BDD at dq and $-dq$.
5. Verify every output, and return the shortest verified nonzero lattice vector over all guesses and starts.

Theorem 10.1 (Conditional main theorem). *Assume Hypothesis 9.2. Algorithm 1 solves search-SVP with probability at least $2/3$ in expected time and worst-case space*

$$2^{n/2+o(n)},$$

up to factors polynomial in the input length.

Proof. Choose the correct length guess d . By Proposition 9.3, one start snaps to a shortest parity with probability at least $2^{-\kappa(n)} - 2^{-\Omega(n)}$. Taking $K = \Theta(2^{\kappa(n)}) = 2^{o(n)}$ independent Haar starts gives constant probability of at least one correct snap. Condition on this event and call its parity θ . By Lemma 4.3, the target point is exactly $v/2 \bmod L$ for a shortest vector v . The target estimator with $n^C 2^{2rn}$ samples has operator error $O(\mu_{r,d}/n)$ by the same concentration argument as Hhan's Lemma 3.5. Theorem 2.3 makes one of the BDD queries return $\pm v$. Exact verification prevents any false candidate from affecting the output. Standard independent repetitions raise the success probability to $2/3$.

The BDD preprocessing and one above-smoothing DGS call cost $2^{n/2+o(n)}$ time and space. A smooth field/gradient query costs

$$2^{(2t+\iota(t,r)+o(1))n} = 2^{(0.4001086+o(1))n}.$$

There are $SK = 2^{o(n)}$ such queries. Each target midpoint query costs

$$2^{(2r+o(1))n} = 2^{(0.4735172+o(1))n},$$

and there are $K = 2^{o(n)}$ of them. All remaining linear algebra, wrapping, rounding, verification, and BDD query work has a subexponential multiplicative overhead. Thus preprocessing dominates at exponent $1/2$. The sample panel and BDD preprocessing data fit in $2^{n/2+o(n)}$ space. $\square$

The exponent ledger also gives a quantitatively weaker sufficient basin bound.

Corollary 10.2 (Exponential basin allowance). *Suppose (33) is replaced by*

$$p_{L,d}(n^{-C_1}) \geq 2^{-(\delta+o(1))n}.$$

Algorithm 1 still runs in $2^{n/2+o(n)}$ time whenever

$$\delta < \min \left\{ \frac{1}{2} - (2t + \iota(t, r)), \frac{1}{2} - 2r \right\} = 0.0264828 \dots \quad (34)$$

Proof. Use $K = 2^{(\delta+o(1))n}$ starts. The smooth and target work exponents become $2t + \iota + \delta$ and $2r + \delta$, respectively. Both remain below $1/2$ under (34). $\square$

11 Foundational consequences

We collect several consequences that are implicit in the Fourier representation but are not needed in Hhan’s discrete Walsh formulation.

11.1 Information lives on the whole torus

The samples used for one parity Hessian are not tied to that parity. They encode the entire trigonometric matrix field (11). Evaluating at a new point does not require new randomness; it requires only another deterministic pass over the same samples. The same is true for every fixed derivative order. This converts parity search from a sample-access problem into a geometric query problem.

11.2 Scale transport is needed only for the observable

The narrowing identity transports the Hessian observable from r to t . It does not transport particles between affine cosets, nor does it move a midpoint branch. Lemma 4.2 shows that the parity half-grid is fixed at all temperatures, and Lemma 4.3 transfers the discrete address exactly. The final rank-one certificate is always evaluated at the original target scale $r > t_0$.

11.3 No eigengap is needed during search

The target midpoint needs a leading eigendirection, and Hhan proves the required gap there. The search trajectory need not have one. The soft spectral density P_T in (30) supplies a canonical direction through collisions. Thus the geometric hypothesis concerns basin mass and dynamical margin, not a uniform eigengap over the torus.

11.4 All false candidates are one-sided errors

The algorithm never trusts the value of the soft objective or the output of an out-of-promise BDD query as a certificate. It verifies lattice membership exactly and returns the shortest verified nonzero vector. The only role of the basin hypothesis is to guarantee that at least one query is correct; it is not used to exclude incorrect queries.

12 Finite evidence

The experiments in this section are included only as checks of the formulas and as evidence for the geometric target. They are not used in any proof.

12.1 Soft ascent on a cancellation family

Consider a basis with Gram matrix having diagonal one and a common negative off-diagonal chosen so that the all-ones coefficient vector has length 0.97. In the tested dimensions it is the unique shortest pair, while each basis column has length one. At the rigorously controlled scale $t = 0.159919\dots$, $\gamma = 16$, Fourier cutoff two, and 256 scrambled Sobol starts, the soft law gave the following results.

n	shortest hits	hit fraction	distinct parities	median evaluations
3	65	0.253906	7	14
4	40	0.156250	15	16
5	30	0.117188	31	18
6	25	0.097656	50	19

The maximum half-grid error among terminal points was below $2.9 \cdot 10^{-6}$. The number of discovered parity maxima is already large, but the shortest parity retains visible basin mass on this small, well-conditioned family. Theorem 13.1 shows why such evidence cannot be extrapolated uniformly to anisotropic lattices.

12.2 Finite-population oracle checks

On explicitly enumerated dual cubes, the population identity (18) held to floating-point precision, and every narrowing weight was at most one as predicted by (17). Independent-source experiments showed decreasing value and gradient error with sample size. No finite-dimensional measurement is used to infer an asymptotic basin lower bound.

13 Resolution of the universal-basin conjecture

The universal Haar-seeded law has a product-lattice obstruction. It is not merely that the preceding estimates fail to prove a large basin: the basin claimed in the former Conjecture 13.1 can be exponentially small.

Theorem 13.1 (Anisotropic product obstruction). *Fix $t = 0.16$, $\gamma = n^2$, any polynomial step count, and any deterministic step schedule of polynomial bit complexity as in the soft-Hessian seed law. For all sufficiently large n , let*

$$M = 2^n, \quad L_n = \text{diag}(1, M, \dots, M)\mathbb{Z}^n. \quad (35)$$

For every correct guess $1 \leq d \leq 1 + 1/n$, the exact soft-Hessian seed law reaches a shortest-vector parity with probability at most

$$2^{-n} + 2^{-2n+2} + 4n2^{-4n} \leq 2^{-n+1}. \quad (36)$$

Consequently, for every $\varepsilon > 0$,

$$p_{L_n, d}(\varepsilon) \leq 2^{-n+1}. \quad (37)$$

Hypothesis 9.2, and hence the former universal-basin Conjecture 13.1, is false.

Proof. Write $B = \text{diag}(1, M, \dots, M)$ and use coefficient coordinates $z = Bu$ on the torus. The only shortest vectors of L_n are $\pm e_1$, so the unique shortest parity is $\theta_* = e_1 \bmod 2$.

Let $s = 1/\xi_t(d)$. Since t is fixed and $d \in [1, 1 + 1/n]$, there are absolute constants $c_0, c_1 > 0$ such that

$$c_0 n^{-1/2} \leq s \leq c_1 n^{-1/2}. \quad (38)$$

For $h > 0$, define the one-dimensional periodic Gaussian

$$g_{h,s}(x) := \sum_{m \in \mathbb{Z}} e^{-\pi(x+mh)^2/s^2}, \quad f_{h,s}(x) := \frac{g_{h,s}(x)}{g_{h,s}(0)}.$$

Separation of the lattice sum gives

$$F_t(z) = f_{1,s}(z_1) \prod_{j=2}^n f_{M,s}(z_j). \quad (39)$$

We use the following elementary localization estimate. For fixed $a \in (0, 1/4)$ and $0 \leq k \leq 3$, differentiation of each Gaussian summand shows, uniformly for $0 < s \leq h$,

$$\left| f_{h,s}^{(k)}(x) \right| \leq P_{k,a}(h + s^{-1}) \exp(-c_a h^2/s^2) \quad (40)$$

whenever $\text{dist}(x, h\mathbb{Z}) \geq ah$. Without the distance assumption the same derivative is at most $P_k(h + s^{-1})$. Here the P 's are fixed polynomials and $c_a > 0$. Indeed, the k th derivative of $e^{-\pi y^2/s^2}$ is a degree- k polynomial in y/s^2 times the same Gaussian. Split the sum at a nearest translate of $h\mathbb{Z}$; the nearest term has $|y| \geq ah$, and the two remaining tails are bounded by geometric Gaussian tails. Division is harmless because $g_{h,s}(0) \geq 1$. Finally, $0 < f_{h,s} \leq 1$: Poisson summation expresses its numerator as a Fourier series with nonnegative coefficients, whose absolute value is at most its value at zero.

Apply (40) with the constant $a/2$. Suppose some long coordinate satisfies

$$\text{dist}(u_j, \mathbb{Z}) \geq a/2, \quad j \geq 2. \quad (41)$$

Then $\text{dist}(z_j, M\mathbb{Z}) \geq aM/2$. Every third partial derivative of the product (39) contains either $f_{M,s}(z_j)$ or one of its first three derivatives. The other undifferentiated factors are at most one, and at most three other

factors are differentiated. Equations (38)–(40) therefore give a fixed polynomial P and a constant $c > 0$ such that

$$\max_{|\alpha|=3} |\partial^\alpha F_t(z)| \leq P(n, M)e^{-cnM^2}. \quad (42)$$

The derivative of the traceless Hessian is a linear combination of these third derivatives. Moreover, by (30),

$$|\partial_j \Psi(z)| = |\text{tr}(P_{\mathcal{T}_t(z)} \partial_j \mathcal{T}_t(z))| \leq \|\partial_j \mathcal{T}_t(z)\|_{\text{op}}, \quad (43)$$

because $P_{\mathcal{T}_t(z)}$ is positive semidefinite with trace one. Thus no factor involving $1/\sigma_{t,d}$ is lost. From (31), (42), and (43), after enlarging P ,

$$\|V(z)\| \leq P(n, M)e^{-cnM^2} \quad (44)$$

whenever (41) holds.

Let $E_n = \sum_{s=0}^{S-1} |\eta_s|$. On the input (35), polynomially many step sizes of polynomial bit complexity satisfy $E_n \leq 2^{q(n)}$ for some fixed polynomial q . Since $M = 2^n$, (44) implies for all sufficiently large n that

$$E_n P(n, M)e^{-cnM^2} < \tau_n, \quad \tau_n := 2^{-4n}. \quad (45)$$

Set $a = 1/8$ and define the exceptional active slabs

$$A := \{u \in [0, 1)^n : \text{dist}(u_j, \mathbb{Z}) < a \text{ for every } j \geq 2\}.$$

Their Haar measure is

$$\text{Haar}(A) = (2a)^{n-1} = 2^{-2n+2}. \quad (46)$$

If $u_0 \notin A$, choose a long coordinate satisfying (41). The coefficient displacement in one step is $B^{-1}\eta_s V(Bu_s)$, whose norm is at most $|\eta_s| \|V(Bu_s)\|$ because $\|B^{-1}\|_{\text{op}} = 1$. Equations (44)–(45), applied inductively, show that the total coefficient displacement is below $\tau_n < a/2$. The witness coordinate therefore stays at distance at least $a/2$ from $\mathbb{Z}$, so the localization estimate remains valid at every step.

The coordinate rounding map $u \mapsto \text{round}(2u) \bmod 2$ changes only across $1/4$ and $3/4$ in each circle coordinate. Let C be the set of coefficient points within τ_n of one of these boundaries in at least one coordinate. A union bound gives

$$\text{Haar}(C) \leq 4n\tau_n. \quad (47)$$

Outside $A \cup C$, the final parity equals the initial parity. The initial parity of a Haar point is uniform on $\mathbb{F}_2^n$. Hence

$$\Pr[\text{final parity} = \theta_*] \leq \Pr[\text{initial parity} = \theta_*] + \text{Haar}(A) + \text{Haar}(C),$$

which is (36).

Finally, every seed in the robust set $\mathcal{S}_{L_n, d}(\varepsilon)$ must succeed for the exact vector field itself, since that field is one of the allowed perturbations. The robust set is therefore contained in the exact success set. This proves (37). Since $n - 1$ is not $o(n)$, no universal $\kappa(n) = o(n)$ can satisfy (33). $\square$

13.1 An exact lattice-adapted parity sieve

The product obstruction contains parity information that a Haar law ignores. Let $v = Bz$ be shortest and put $\theta = z \bmod 2$. For $k \in \mathbb{Z}^n$, write $y_k = B^{-T}k \in L^*$.

Proposition 13.2 (Dual-syndrome and Hamming sieve). *Suppose (5) holds. If $d\|y_k\| < 1$, then*

$$k^T z = 0, \quad k^T \theta = 0 \pmod{2}. \quad (48)$$

Moreover, with

$$q(B, d) := \min\{n, \lfloor d^2 \|B^{-1}\|_{\text{op}}^2 \rfloor\}, \quad (49)$$

every shortest parity is nonzero and has Hamming weight at most $q(B, d)$. Consequently, for any matrix K of verified rows satisfying (48), every shortest parity lies in the explicitly defined set

$$\mathcal{P}_{B, d, K} := \{\theta \in \mathbb{F}_2^n \setminus \{0\} : K\theta = 0, \text{ wt}(\theta) \leq q(B, d)\}, \quad (50)$$

whose cardinality is at most

$$Q_q := \sum_{j=1}^{q(B,d)} \binom{n}{j}. \quad (51)$$

It can be enumerated by filtering either the Hamming ball or the binary kernel, in time polynomial in

$$E_{q,K} := \min\{Q_q, 2^{n - \text{rank}_{\mathbb{F}_2} K}\}. \quad (51a)$$

Proof. The integer $k^T z = \langle y_k, v \rangle$ has absolute value at most $d\|y_k\| < 1$, proving (48). Also

$$\|z\| \leq \|B^{-1}\|_{\text{op}} \|v\| \leq d\|B^{-1}\|_{\text{op}}.$$

Every odd coordinate of z contributes at least one to $\|z\|^2$, which proves (49)–(51). Finally, a shortest parity cannot be zero: if $z = 2w$, then the nonzero lattice vector $Bw = v/2$ is shorter. $\square$

Corollary 13.3 (Sparse-parity solvable class). *If $E_{q,K} = 2^{o(n)}$, enumerating (50), applying the target midpoint estimator and BDD endpoint to every retained parity, and verifying the outputs solves SVP in $2^{n/2+o(n)}$ time and space. More generally it suffices that $E_{q,K} \leq 2^{(1/2+o(1))n}$ and*

$$\frac{1}{n} \log_2 |\mathcal{P}_{B,d,K}| < 0.0264828\dots \quad (52)$$

In particular, $q = o(n)$ suffices. If $q/n \rightarrow \rho$, it suffices that $H_2(\rho) < 0.0264828\dots$, where $H_2(\rho) = -\rho \log_2 \rho - (1-\rho) \log_2 (1-\rho)$, equivalently $\rho < 0.00264782\dots$

Proof. Every candidate midpoint query costs $2^{(2r+o(1))n}$ operations using the common target sample panel, while the filtering work is polynomial in $E_{q,K}$. Combine this with (34); (51) has exponent $H_2(\rho) + o(1)$ when $q/n \rightarrow \rho$. $\square$

For the lattice (35), the rows $k = e_2, \dots, e_n$ have $\|y_k\| = 2^{-n}$. They force every long parity bit to zero. The only remaining nonzero parity is e_1 , so the counterexample to universal Haar ascent is solved deterministically by the input geometry.

13.2 A spectral support-chain cover

The sieve can be vacuous when a shortest vector is a dense coefficient cancellation. The support-chain remedy is to transport the low-energy coarse response and store only its sparse rounding detail. Put $G = B^T B$. Let $U \subset \mathbb{R}^n$ have dimension k and orthogonal projector P_U , and suppose every point of the coefficient ellipsoid

$$\mathcal{E}_d := \{z \in \mathbb{R}^n : z^T G z \leq d^2\} \quad (53)$$

satisfies

$$\|z\| \leq R, \quad \|(I - P_U)z\| \leq C. \quad (54)$$

For the first k Gram eigendirections one may take $R = d/\sqrt{\lambda_1(G)}$ and $C = d/\sqrt{\lambda_{k+1}(G)}$.

Theorem 13.4 (Spectral-tube parity cover). *Let $N = n(2\lceil R \rceil + 2)$ and $s = \min\{n, \lfloor 4C^2 \rfloor\}$, $m = \lceil C + 1/2 \rceil$. Every shortest parity is contained in an explicitly enumerable cover of cardinality at most*

$$2^k \left(\sum_{j=0}^k \binom{N}{j} \right) \left(\sum_{i=0}^s \binom{n}{i} (2m)^i \right). \quad (55)$$

In particular the cover has size $2^{o(n)}$ whenever, uniformly in the input model,

$$k \log(n(R+1)) + C^2 \log(n(C+1)) = o(n), \quad (56)$$

and the displayed catalog has only a polynomial factor in the input length.

Proof. The coordinate half-integer hyperplanes meeting the radius- R ball in U number at most N . Their k -dimensional arrangement, including all tie faces, has at most $2^k \sum_{j=0}^k \binom{N}{j}$ rounding labels.

For an integer $z \in \mathcal{E}_d$, put $x = P_U z$, $u = \text{round}(x)$, and $w = z - u$. If $w_i \neq 0$, then z_i is not the nearest integer to x_i , so $|z_i - x_i| \geq 1/2$. Equation (54) implies

$$\text{wt}(w) \leq 4C^2, \quad |w_i| \leq C + 1/2.$$

Thus $z \bmod 2$ is obtained from one arrangement label by adding one of the integer details counted by the second factor in (55). This proves the cover and (56). Rational spectral projectors may be certified by enlarging R, C ; enumeration is output-sensitive, so an enormous R is not treated as polynomial in $\log R$. $\square$

The simplex-cancellation family from Section 12 has a one-dimensional minimum Gram eigenspace, $R = \Theta(\sqrt{n})$, and $C < 1$. Hence (55) is polynomial even though its shortest parity is the all-ones word and the Hamming sieve is vacuous. The rectangular family also has $k = 1$ with $R = O(1)$ and $C = 2^{-\Omega(n)}$. The same stable lifting construction thus handles both known obstructions.

13.3 Exact factor transports

The size of the parity catalog is not the only exact complexity measure. The coefficient energy itself may have a narrow factorization. Let $G = B^T B$ and, for the length guess $d \geq \lambda$, put

$$R_i := \lfloor d \|\text{row}_i(B^{-1})\|_2 \rfloor, \quad D_i := \{-R_i, \dots, R_i\}. \quad (57)$$

Join i and j when $G_{ij} \neq 0$. For an elimination order π , let $K_i(\pi)$ contain the variable eliminated at step i and its remaining neighbors, including earlier fill edges, and define

$$W_\pi(B, d) := \max_i \sum_{j \in K_i(\pi)} \log_2(2R_j + 1). \quad (58)$$

Theorem 13.5 (Weighted Gram–GDL elimination). *For every explicitly supplied order π , exact search-SVP is solvable in space*

$$2^{W_\pi(B, d)} \text{poly}(n, \text{size}(B), \log(d + 1))$$

and at most n times this quantity in time. The algorithm returns a minimizing coefficient vector by a backward-preimage pass.

Proof. If $v = Bz$ and $\|v\| \leq d$, then

$$|z_i| = |\langle \text{row}_i(B^{-1}), v \rangle| \leq d \|\text{row}_i(B^{-1})\|_2,$$

so every shortest coefficient vector lies in $\prod_i D_i$. On this finite product,

$$\|Bz\|^2 = \sum_i G_{ii} z_i^2 + \sum_{i < j} 2G_{ij} z_i z_j \quad (59)$$

is a unary-and-pairwise factor graph with exactly the displayed Gram interactions. Exact min–sum elimination in order π forms a table only on the current bag $K_i(\pi)$, hence at most $2^{W_\pi(B, d)}$ entries. Store an attaining value of the eliminated variable at every table entry; the reverse pass recovers a global minimizer. Exact rational arithmetic has polynomial bit growth in these operations.

To exclude zero, run the elimination once for each $p \in [n]$ with the domain $D_p \setminus \{0\}$ and take the best answer. Every nonzero assignment appears in at least one run, and a shortest assignment lies in the bounded product. This proves the claim. $\square$

The same proof applies to auxiliary exact factorizations. Suppose finite variables x have domains E_x , local energy factors and zero–infinity constraint factors represent (59), and projection of the finite-objective assignments onto z is exactly $\prod_i D_i$. A tree decomposition $\mathcal{D}$ has weighted junction width

$$J(\mathcal{D}) := \max_{A \in \mathcal{D}} \sum_{x \in A} \log_2 |E_x|. \quad (60)$$

Min-sum and backward preimage then cost $2^{J(\mathcal{D})}$ poly up to the same n zero-exclusion runs. For example, after clearing denominators, a row $(Bz)_k = \sum_i B_{ki} z_i$ can be represented by a binary tree of exact partial sums and one unary square factor. A sum over S needs only the integer interval

$$\left[-\sum_{i \in S} |B_{ki}| R_i, \sum_{i \in S} |B_{ki}| R_i \right]. \quad (61)$$

Thus a dense Gram graph need not imply a wide exact computation. Conversely the state intervals in (61) can be expensive; the certified quantity is the weighted junction width, not unweighted treewidth.

Corollary 13.6 (Exact-factor escape). *If an efficiently constructed order or auxiliary decomposition has width at most $(1/2 + o(1))n$, it gives an unconditional half-exponential algorithm on that input. In particular, width below $(0.0264828 - \varepsilon)n$ disposes of the input well inside the later parity-query allowance. No optimization over undisplayed bases or decompositions is assumed.*

There is also an exact dense-factor escape. It is classical in the language of lattices of Voronoi's first kind [7]; the short proof is included because its coarea form is the relevant transport principle.

Theorem 13.7 (Obtuse-superbase minimum cut). *Let $b_1, \dots, b_{n+1}$ be a superbase: $b_1, \dots, b_n$ is a lattice basis, $\sum_i b_i = 0$, and $\langle b_i, b_j \rangle \leq 0$ for $i \neq j$. Put $w_{ij} = -\langle b_i, b_j \rangle \geq 0$. A global minimum nontrivial cut of the graph with weights w_{ij} returns a globally shortest lattice vector.*

Proof. Coefficients $c_i \in \mathbb{Z}$ are defined modulo addition of a common integer, and the graph-Laplacian identity gives

$$\left\| \sum_i c_i b_i \right\|^2 = \sum_{i < j} w_{ij} (c_i - c_j)^2. \quad (62)$$

For a nonzero vector normalize $\min_i c_i = 0$ and put $S_t = \{i : c_i \geq t\}$ for $1 \leq t \leq \max_i c_i$. Since an integer a satisfies $a^2 \geq |a|$,

$$\left\| \sum_i c_i b_i \right\|^2 \geq \sum_t \sum_{i < j} w_{ij} |\mathbf{1}_{i \in S_t} - \mathbf{1}_{j \in S_t}|. \quad (63)$$

Every summand in t is a nontrivial cut capacity, so (63) is at least the global minimum cut. Conversely, the subset vector $\sum_{i \in S} b_i$ has squared norm exactly the capacity of the cut S . The two minima are equal, and a minimum cut supplies the desired vector. $\square$

Obtuseness of a displayed rational superbase is exactly checkable and a global minimum cut is polynomial-time computable. In particular the simplex-cancellation family used above has an obtuse displayed superbase: its exponential subset fan is submodular and is not part of the remaining hard core.

13.4 A direct gap escape

Fix an oriented shortest vector v , write $a = v/\lambda$, and let

$$\gamma(v) := \min\{\|w\|/\lambda : w \in L, w \notin \mathbb{R}v\}. \quad (64)$$

Positive multiples of v cannot win the first-exit competition before v . Define

$$a_\gamma := \begin{cases} \sqrt{1 - \gamma^2/4}, & 1 \leq \gamma \leq \sqrt{2}, \\ 1/\gamma, & \gamma \geq \sqrt{2}. \end{cases} \quad (65)$$

Lemma 13.8 (Gap-dependent winner cap). *If $u \in S^{n-1}$ satisfies $\langle u, a \rangle \geq a_\gamma$, the first facet met by the ray $\mathbb{R}_{\geq 0} u$ is F_v . Consequently the two winner caps for $\pm v$ have total mass*

$$2^{-(\delta_{\text{gap}}(\gamma) + o(1))n}, \quad \delta_{\text{gap}}(\gamma) = -\frac{1}{2} \log_2(1 - a_\gamma^2). \quad (66)$$

Proof. Write a noncollinear competitor as $w = q\lambda b$, with $\|b\| = 1$ and $q \geq \gamma$. Minimality of $v - w$ implies $\langle a, b \rangle \leq q/2$. Put $s = \langle u, a \rangle$. For $q \leq \sqrt{2}$, maximizing $\langle u, b \rangle$ under this constraint shows that $\langle u, b \rangle/q \leq s$ once $s \geq \sqrt{1 - q^2/4}$. For $q \geq \sqrt{2}$, the bound $\langle u, b \rangle \leq 1$ gives the sufficient condition $s \geq 1/q$. Both thresholds decrease in q , yielding (65). The fixed-height spherical-cap asymptotic is $(1 - a_\gamma^2)^{n/2}$ up to subexponential factors. $\square$

At the query allowance in (34), this lemma closes the ledger whenever

$$\gamma > (1 - 2^{-2(0.0264828\dots)})^{-1/2} = 5.2670066\dots \quad (67)$$

The second branch in (65) is essential; continuing the first branch beyond $\sqrt{2}$ would give an invalid constant.

13.5 Causal transport through the Voronoi tiling

Let V be the origin Voronoi cell and let Q be nearest-lattice decoding away from its measure-zero ties. For an oriented Voronoi-relevant vector w , write

$$F_w := V \cap \{x : \langle x, w \rangle = \|w\|^2/2\}, \quad A_w := \mathcal{H}^{n-1}(F_w). \quad (68)$$

For a parity θ , let A_θ sum these areas over the facets whose coefficient vectors have parity θ . Draw X uniformly from V and U uniformly from S^{n-1} , independently, and set

$$C_\eta := Q(X + \eta\lambda U) - Q(X). \quad (69)$$

Theorem 13.9 (Causal tiling kernel and transport conservation). *For every nonzero parity θ ,*

$$\lim_{\eta \rightarrow 0+} \frac{\Pr[C_\eta \bmod 2L = \theta]}{\eta} = d_n \frac{\lambda A_\theta}{\det L}, \quad d_n := \frac{\Gamma(n/2)}{2\sqrt{\pi} \Gamma((n+1)/2)} = \Theta(n^{-1/2}). \quad (70)$$

Moreover, with

$$q_w := \frac{\|w\| A_w}{2n \det L}, \quad (71)$$

where both orientations are included,

$$\sum_w q_w = 1, \quad \frac{1}{2 \det L} \sum_w A_w \frac{ww^T}{\|w\|} = I_n, \quad \sum_w q_w \frac{ww^T}{\|w\|^2} = \frac{1}{n} I_n. \quad (72)$$

Proof. Outside a tube of volume $O(\eta^2)$ around the codimension-two faces, a segment of length $\eta\lambda$ crosses at most one facet. At a regular point of F_w , the starting points in V which cross outward have normal thickness

$$\eta\lambda (\langle U, w/\|w\| \rangle)_+ + O(\eta^2).$$

Integrate over F_w , average over U , divide by $\text{vol } V = \det L$, and use $\mathbb{E}(U_1)_+ = d_n$. Summing the facets of parity θ proves (70).

The pyramids $\text{conv}(0, F_w)$ partition V up to measure zero; their volumes are $\|w\| A_w/(2n)$. This proves the first identity in (72). For the matrix identity, put $R(x) = x - Q(x)$. Haar translation invariance on $\mathbb{R}^n/L$ gives

$$\mathbb{E}[Q(X + h) - Q(X)] = h - \mathbb{E}[R(X + h) - R(X)] = h. \quad (73)$$

Differentiate at $h = 0$. Crossing F_w has one-sided rate $A_w(\langle h, w/\|w\| \rangle)_+/\det L$ and transports the label by w . Pairing w and $-w$ converts the positive parts into the middle identity of (72). Multiplying each summand by the definition (71) gives the last identity; taking the trace also recovers $\sum_w q_w = 1$. $\square$

This realizes the principle that information lives in transport: the label differences across actual first-arrival cells form an exact isotropic frame. If $q_{\min}$ is the sum of (71) over the oriented shortest facets, (70) gives total shortest-parity derivative at least

$$2nd_n q_{\min} = \Theta(\sqrt{n}) q_{\min}. \quad (74)$$

One may draw X uniformly in any fundamental parallelepiped and subtract the nearest labels before and after the step; translation covariance gives the same law. This is a geometric representation, not an available CVP oracle.

The isotropic identity by itself cannot lower-bound $q_{\min}$.

Proposition 13.10 (Isotropy alone has no atom bound). *For every $n \geq 2$ and every $0 < \varepsilon < 1/n$, there is a finite centrally symmetric probability frame on S^{n-1} with covariance I_n/n but with total weight exactly ε on a designated antipodal pair.*

Proof. Fix a unit vector u , an orthonormal basis $e_2, \dots, e_n$ of $u^\perp$, and put

$$\alpha = \frac{1/n - \varepsilon}{1 - \varepsilon}.$$

Give $\pm u$ total weight ε . Distribute the remaining weight uniformly over the centrally symmetric vectors

$$\pm(\sqrt{\alpha}u \pm \sqrt{1 - \alpha}e_j), \quad 2 \leq j \leq n.$$

The cross terms cancel. The covariance in the u direction is $\varepsilon + (1 - \varepsilon)\alpha = 1/n$ and that on $u^\perp$ is $(1 - \varepsilon)(1 - \alpha)/(n - 1) = 1/n$. $\square$

Thus a proof of the remaining theorem must use the fact that (72) is the cone-volume measure of a lattice Voronoi parallelohedron, or must combine it with the exact escapes above; frame conservation alone cannot suffice.

There is a second obstruction to any proof that uses only local shortest geometry. Let p_w be the spherical measure of rays exiting V through F_w , and let $p_{\min}$ sum these weights over the oriented shortest facets.

Proposition 13.11 (Kissing perturbation stress test). *Suppose a lattice L has K oriented shortest vectors. For every $\zeta > 0$ there is an arbitrarily small linear perturbation L' for which one antipodal pair is uniquely shortest and*

$$p_{\min}(L') + q_{\min}(L') \leq \frac{4 + \zeta}{K}.$$

If L is rational, the perturbation may be chosen rational.

Proof. Group the K oriented shortest vectors into $K/2$ antipodal pairs. Their total direct-exit mass is at most one and their total cone-volume mass is at most one. Some pair $\{\pm v\}$ therefore has combined p - and q -weight at most $4/K$.

Fix a shortest vector v , put $a = v/\lambda$, and apply $T_\tau = I - \tau aa^T$ for sufficiently small $\tau > 0$. Among the old shortest vectors, $\pm v$ shrink strictly more than every other pair, by the strict case of Cauchy–Schwarz. Lattice discreteness gives a positive gap to the next length shell, so for small enough τ no longer vector overtakes them. Thus $\pm T_\tau v$ are uniquely shortest.

Every shortest facet contains the inball proved below, hence persists under a sufficiently small perturbation. Its solid angle, area, support distance, and the Voronoi-cell volume vary continuously. The two oriented weights of the new shortest pair therefore converge to their two old weights as $\tau \rightarrow 0$, giving the stated bound. When L is rational, $aa^T = vv^T/\lambda^2$ is rational, so rational τ preserves rationality. $\square$

Consequently any exponential kissing family with exponent κ would rule out a universal cone-volume bound with $\delta < \kappa$, unless its small perturbations are removed by one of the exact joint-core escapes. This is why code-derived high-kissing configurations are a necessary stress test for the remaining theorem; a spherical-code bound alone does not decide whether the corresponding lattice inputs survive those escapes.

13.6 A log-theta collision kernel

The causal crossing need not occur near the center $v/2$ of its facet. This prevents direct use of the raw theta Hessian: its two nearest terms retain a tangential rank-one component. The logarithmic Hessian cancels that component exactly. Put

$$Z_s(x) := \rho_s(L + x) = \sum_{w \in L} e^{-\pi \|x - w\|^2 / s^2}$$

and define the posterior on lattice sites

$$\Pr_x[W = w] := \frac{e^{-\pi\|x-w\|^2/s^2}}{Z_s(x)}. \quad (75)$$

Theorem 13.12 (Log-theta covariance and two-site stability). *For every $x \in \mathbb{R}^n$,*

$$\mathcal{C}_s(x) := \frac{s^4}{4\pi^2} \left(\nabla^2 \log Z_s(x) + \frac{2\pi}{s^2} I_n \right) = \text{Cov}_x(W). \quad (76)$$

Let $x \in F_v$ for a shortest vector v , set $a = e^{-\pi\|x\|^2/s^2} = e^{-\pi\|x-v\|^2/s^2}$, and define

$$\epsilon_0(x) := \frac{1}{2a} \sum_{w \notin \{0,v\}} e^{-\pi\|x-w\|^2/s^2}, \quad \epsilon_2(x) := \frac{1}{2a\lambda^2} \sum_{w \notin \{0,v\}} \|w\|^2 e^{-\pi\|x-w\|^2/s^2}. \quad (77)$$

If $\epsilon_0(x), \epsilon_2(x) \leq \epsilon \leq 1$, then

$$\|\mathcal{C}_s(x) - \frac{1}{4}vv^T\|_{\text{op}} \leq 6\epsilon\lambda^2. \quad (78)$$

Consequently, when $\epsilon = o(1)$ the leading eigendirection of $\mathcal{C}_s(x)$ is $o(1)$ -close to $\mathbb{R}v$.

Proof. Differentiating the log partition function gives

$$\nabla \log Z_s(x) = -\frac{2\pi}{s^2}(x - \mathbb{E}_x W)$$

and a second differentiation gives (76).

For (78), divide all lattice sites by λ and put $u = v/\lambda$. Relative to the total weight $2a$ of $\{0, v\}$, let r_1 and r_2 be the first and second raw moments of the remaining sites. Cauchy–Schwarz gives $\|r_1\| \leq \sqrt{\epsilon_0\epsilon_2}$ and $\|r_2\|_{\text{op}} \leq \epsilon_2$. If $m = \mathbb{E}_x(W/\lambda)$, then

$$\|m - u/2\| \leq \sqrt{\epsilon_0\epsilon_2} + \epsilon_0/2 \leq 3\epsilon/2,$$

while the normalized second raw moment differs from $uu^T/2$ by at most $\epsilon_2 + \epsilon_0/2$. Subtracting the two outer products bounds the covariance error by

$$\epsilon_2 + \epsilon_0/2 + 3\epsilon/2 + (3\epsilon/2)^2 \leq 6\epsilon.$$

Rescaling by λ^2 proves (78). Standard eigenvector perturbation then proves the last assertion. $\square$

There is an exact line-scan interpretation. Suppose $x(t) = x_0 + t\xi$ crosses F_v at $t = t_0$. Ignoring the remainder in (77), the posterior probability of the site v is

$$p(t) = \left(1 + \exp \left[-\frac{2\pi\langle \xi, v \rangle (t - t_0)}{s^2} \right] \right)^{-1}, \quad (79)$$

and

$$\mathcal{C}_s(x(t)) = p(t)(1 - p(t))vv^T. \quad (80)$$

Thus its leading eigenvalue has a logistic ridge centered exactly at the facet, of physical width $s^2/|\langle \xi, v \rangle|$. The proof above, with the two distinguished weights no longer assumed equal, shows that (80) has operator error at most $10\epsilon\lambda^2$ whenever the analogues of ϵ_0, ϵ_2 relative to their combined weight are at most ϵ . If the crossing angle and ϵ^{-1} are polynomial and $s^2/\lambda^2 = 1/\text{poly}(n)$, a polynomial grid resolves this ridge and its eigendirection.

The quantity $\mathcal{C}_s$ is algebraically obtainable from $Z_s, \nabla Z_s, \nabla^2 Z_s$, all encoded by the common DGS Fourier panel. However, this normalization converts the existing absolute empirical error into relative error. The analytic part of the remaining theorem is therefore reduced to two explicit estimates on a robust shortest-facet core:

$$\epsilon_0 + \epsilon_2 \leq n^{-\Omega(1)} \quad \text{and} \quad Z_s(x) \text{ is above the empirical relative-error floor.} \quad (81)$$

Neither estimate follows from nearest-pair geometry alone.

13.7 Robust shortest-facet cores

Lemma 13.13 (Shortest-facet robust core). *Every shortest facet F_v contains, in its affine hyperplane, the ball of radius $\lambda/(2\sqrt{3})$ centered at $v/2$. For $0 < \varepsilon < 1$, the homothetic core*

$$F_v(\varepsilon) := (1 - \varepsilon)F_v + \varepsilon(v/2) \quad (82)$$

has area $(1 - \varepsilon)^{n-1}A_v$, is at tangential distance at least $\varepsilon\lambda/(2\sqrt{3})$ from every ridge, and every x in the core satisfies

$$\|x - w\|^2 - \|x\|^2 \geq \varepsilon\lambda^2/2 \quad (83)$$

for all $w \in L \setminus \{0, v\}$.

Proof. On the hyperplane of F_v , the distance from $v/2$ to the constraint induced by w is

$$\frac{\|w\|^2 - \langle v, w \rangle}{2\|P_{v^\perp} w\|}. \quad (84)$$

Scale $\lambda = 1$, set $A = \|w\|^2$ and $b = \langle v, w \rangle$. Minimality and Cauchy–Schwarz give $A \geq 1$, $b \leq A/2$, and $b^2 \leq A$. On this domain

$$3(A - b)^2 \geq A - b^2. \quad (85)$$

Indeed, for $1 \leq A \leq 4$ the difference is minimized at $b = A/2$, where it equals $A(A - 1)$; for $A \geq 4$ it is minimized at $b = \sqrt{A}$. Equations (84)–(85) give the inball.

Every defining slack of F_v is affine and nonnegative on F_v . At $v/2$ the slack belonging to $w \neq 0, v$ is $\|w\|^2 - \langle v, w \rangle \geq \|w\|^2/2 \geq \lambda^2/2$. The convex combination (82) therefore gives (83), while applying it to the inball gives the ridge distance. Homothety scales $(n - 1)$ -area by $(1 - \varepsilon)^{n-1}$. $\square$

Taking $\varepsilon = 1/n$ retains a constant fraction of every shortest-facet area and gives inverse-polynomial positional reach and energy margin. Thus robustification costs no exponential factor. The inball also yields the following unconditional, though generally weak, density escape. If $N_{\min}$ counts oriented shortest facets and κ_m is the volume of the unit m -ball, then

$$q_{\min} \geq \frac{N_{\min} \kappa_{n-1} \lambda^n}{2n(2\sqrt{3})^{n-1} \det L}. \quad (86)$$

13.8 Tail suppression and the response-floor barrier

The robust core gives a clean weighted-tail estimate, but its parameter tradeoff is incompatible with the desired mass. We state the obstruction for Hhan’s normalization (6).

Lemma 13.14 (Homothetic weighted-tail bound). *Fix $t > 0$ and a constant $0 < \eta \leq 1$, put $s = 1/\xi_t(d)$, and let $x \in F_v(\eta)$. The tail quantities in (77) satisfy*

$$\epsilon_0(x) + \epsilon_2(x) \leq 2^{(\frac{1}{2} \log_2(t_0/(\eta t)) + o(1))n}. \quad (87)$$

Consequently the universal shell estimate certifies polynomial (indeed, exponential) suppression for fixed parameters when $\eta t > t_0$. Since the available target scale has $t < 1/4$, this forces $\eta > 4t_0 = 0.9258\dots$ and retains at most

$$(1 - 4t_0 + o(1))^{n-1} A_v = 2^{-(3.7540 + o(1))n} A_v. \quad (88)$$

Proof. For $w \in L$, write

$$D_w(z) := \|z - w\|^2 - \|z\|^2 = \|w\|^2 - 2\langle z, w \rangle.$$

This is affine in z . It is nonnegative on F_v for every $w \notin \{0, v\}$, while at the midpoint

$$D_w(v/2) = \|w\|^2 - \langle v, w \rangle = \frac{\|w\|^2 + \|w - v\|^2 - \lambda^2}{2} \geq \frac{\|w\|^2}{2}.$$

The homothetic definition therefore gives $D_w(x) \geq \eta \|w\|^2/2$. Dividing every non-colliding term by the two-site weight $2a$ in (77) yields

$$\epsilon_0(x) + \epsilon_2(x) \leq \frac{1}{2} \sum_{w \neq 0} \left(1 + \frac{\|w\|^2}{\lambda^2}\right) \exp\left(-\frac{\pi\eta\|w\|^2}{2s^2}\right).$$

Apply Corollary 3.2 of [4] with Gaussian dilation $a_\eta = \sqrt{2/\eta}$ and moment degree two. Its exponent is $\frac{1}{2} \log_2(a_\eta^2 t_0/(2t)) = \frac{1}{2} \log_2(t_0/(\eta t))$, which proves (87). The area formula in Lemma 13.13 and $t < 1/4$ prove (88). $\square$

The preceding loss concerns one convenient core. The next bound applies to an arbitrary subset of a shortest facet and exposes the underlying conflict. A normalized theta response estimated from $M = 2^{(m+o(1))n}$ bounded Fourier samples has the usual absolute scale $M^{-1/2} = 2^{-(m/2+o(1))n}$. In fact, for $X \sim D_{L^*,1/s}$ the direct empirical Fourier summand obeys

$$\text{Var}(\cos(2\pi\langle X, x \rangle)) = \frac{1 + F_s(2x)}{2} - F_s(x)^2.$$

Thus its variance is bounded below by a constant whenever $F_s(x) = o(1)$; the $M^{-1/2}$ scale is present rather than merely an artifact of a uniform upper bound. We deliberately allow arbitrary polynomial slack around it.

Theorem 13.15 (Empirical response-floor barrier). *Fix $t > 0$, let $M = 2^{(m+o(1))n}$, and put $\alpha = m/2$ and $s = 1/\xi_t(d)$. For a shortest facet F_v , let $E_v \subseteq F_v$ be any set on which, for fixed $c, C > 0$,*

$$\epsilon_0(x) \leq n^{-c}, \quad F_s(x) \geq n^{-C} 2^{-\alpha n}. \quad (89)$$

Writing $x = v/2 + y$ with $y \perp v$, every $x \in E_v$ satisfies

$$\frac{\|y\|^2}{\lambda^2} \leq \frac{1}{4} \left(\frac{\alpha}{t} - 1\right) + O\left(\frac{\log n}{n}\right). \quad (90)$$

If the leading term on the right lies in $(0, 1/12)$, then

$$\frac{\mathcal{H}^{n-1}(E_v)}{A_v} \leq \left(\sqrt{3(\alpha/t - 1) + O(\log n/n)}\right)^{n-1}. \quad (91)$$

If $\alpha \leq t$, the fraction is instead at most $2^{-\frac{1}{2}n \log_2 n + O(n \log \log n)}$ (and can be empty).

Proof. On F_v , orthogonality gives equal distinguished weights

$$a = \exp\left[-\pi\xi_t(d)^2 \left(\frac{\lambda^2}{4} + \|y\|^2\right)\right].$$

The first condition in (89) implies $Z_s(x) = 2a(1 + \epsilon_0(x)) \leq 3a$ for large n . Since $\rho_s(L) \geq 1$, one has $F_s(x) = Z_s(x)/\rho_s(L) \leq 3a$. Combining this with the response floor in (89), taking base-two logarithms, and using $\xi_t(d)^2 = 4nt \ln 2/(\pi d^2)$ gives

$$t \left(\frac{\lambda^2}{d^2} + \frac{4\|y\|^2}{d^2}\right) \leq \alpha + O(\log n/n).$$

Because $\lambda \leq d \leq (1 + 1/n)\lambda$, this is (90).

Thus E_v lies in an $(n-1)$ -ball about $v/2$ of the radius in (90). Lemma 13.13 lower-bounds A_v by the volume of the $(n-1)$ -ball of radius $\lambda/(2\sqrt{3})$. Dividing the two ball volumes proves (91). When $\alpha \leq t$, (90) gives radius $O(\lambda\sqrt{\log n/n})$; the same volume comparison gives the stated superexponential bound. $\square$

Even granting the entire $2^{n/2+o(n)}$ sample panel to one query gives $m = 1/2$ and $\alpha = 1/4$. Tail suppression from Lemma 13.14 requires $t > t_0$ even at $\eta = 1$. Letting $t \downarrow t_0$ in (91) shows that the accessible fraction of a shortest facet is at most $2^{-(1.0289+o(1))n}$. Conversely, retaining fraction $2^{-(\delta+o(1))n}$ in (91) requires

$$m \geq 2t_0 \left(1 + \frac{2^{-2\delta}}{3}\right) = 0.61169\dots \quad (\delta = 0.0264828), \quad (92)$$

strictly above the half-exponential budget. The actual multi-query ledger is smaller still. Therefore the two estimates in (81) cannot hold on enough cone-volume mass for the proposed empirical log-theta line scan.

13.9 The remaining joint-core theorem

Let $p_{\min}$ denote the spherical measure of directions whose first Voronoi exit has a shortest parity. At allowance $\delta < 0.0264828\dots$, fix an explicit enumerable catalogue of certificates and call an input a *joint core relative to that catalogue* after every certified short-dual quotient and basis preprocessing step if: all Hamming, spectral-tube, and affine parity covers have exponent exceeding δ ; every efficiently displayed exact factor decomposition has cost exceeding the half-exponential budget in Corollary 13.6; and no certified obtuse-superbase representation applies.

This qualification is essential. Without a fixed certificate catalogue, “joint core” is not an intrinsic class of lattices and the proposed inequality is not a well-posed universal assertion. Even after fixing the catalogue, Proposition 13.11 shows that neither isotropy nor local shortest-facet geometry can prove the desired exponent.

The surviving geometric statement is now the following, and is deliberately recorded as a conjecture rather than a theorem.

Conjecture 13.16 (Joint-core cone-volume bound). *For some universal $\delta < 0.0264828\dots$, every joint core satisfies*

$$p_{\min} + q_{\min} \geq 2^{-(\delta+o(1))n}. \quad (93)$$

Theorem 13.9 and Lemma 13.13 settle the geometric conversion from $q_{\min}$ to a robust causal crossing set, losing only a polynomial factor. Proposition 13.10 shows why the matrix conservation law does not prove (93) on its own. Theorem 13.12 supplies the exact line observable and removes all tangential-location dependence. Theorem 13.15, however, proves that even (93) would not instantiate the empirical line scan: the tail and response-floor estimates (81) cannot coexist on the required mass under the stated sample budget. The genuinely viable target is therefore a lower bound on *response-weighted* shortest-collision mass, together with an estimator exploiting its variance, or a different nonlocal observable that avoids division by the exponentially small local theta response. Direct-plus-cone mass alone is no longer a sufficient theorem.

14 Conclusion

Hhan’s midpoint-Hessian theorem already supplies the exact recovery endpoint. The Fourier samples used there contain a larger continuous object: a traceless matrix field and its derivatives on the whole torus. Bounded narrowing importance sampling exposes this field at a rigorously controlled smooth scale with exponent 0.4002, log-trace-exponential smoothing makes its gradient gap-free, and half-grid stationarity turns a successful continuous endpoint into an exact target midpoint without branch transport.

These facts prove a black-box reduction from robust shortest-parity basin mass to exact SVP in $2^{n/2+o(n)}$ time and space. They do not prove that the universal soft-Hessian flow supplies that mass. On the contrary, the anisotropic product family proves an essentially 2^{-n} upper bound for the proposed Haar-seeded law and resolves the former Conjecture 13.1 negatively.

That counterexample does not obstruct every lattice-adapted seed. Exact short-dual equations remove irrelevant parity bits, the coefficient norm gives a sparse Hamming law, and the spectral-tube theorem carries a low-dimensional coarse response together with only a sparse lifting detail. Weighted min–sum elimination gives a different exact escape whenever the coefficient energy has a narrow displayed factorization. An obtuse superbase gives a polynomial minimum-cut algorithm, so the simplex-cancellation family is solved for a stronger reason than its spectral cover.

For inputs surviving those reductions, causal Voronoi transport supplies an exact geometric kernel. Its facet labels carry the crossing parities, its cone-volume weights form an isotropic transport frame, and shortest facets retain robust constant-area cores. A sufficiently large noncollinear gap is also solved by an explicit direct winner cap. What remains is not the former dense spectral core but a response-weighted version of the joint core. Conjecture 13.16 remains a geometric question only after its certificate catalogue is fixed; in any case, it is no longer sufficient for the algorithm. The response-floor barrier proves that the exact log-theta covariance detector cannot combine polynomial tail suppression with the required facet mass in $2^{n/2+o(n)}$ samples. The lower constant in the title is therefore conditional on a different observable or a variance-aware response-weighted collision theorem. The present paper proves the surrounding reduction, exact escapes, and this obstruction, but not that replacement clause.

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